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Smart Learning Environments (2026) 13:6
https://doi.org/10.1186/s40561-025-00423-z

Smart Learning Environments

RESEARCH Open Access

Neuro-symbolic synergy in education: a survey of LLM-knowledge graph integration for explainable reasoning and emotion-aware student support

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Abstract

This article presents a structured survey of recent approaches integrating Large Language Models (LLMs) and Knowledge Graphs (KGs) in education, with a dual focus on explainable reasoning and emotion-aware student support. The objective is to assess how neuro-symbolic architectures and affective computing enhance both transparency and learner well-being in AI-driven tutoring systems. A multimodal literature review was conducted, combining keyword-based searches across Scopus, IEEE Xplore, and SpringerLink from 2020 to 2024, with inclusion criteria focusing on studies addressing LLM explainability, KG integration, and affective adaptation in education. The selected papers were analyzed using a three-axis framework: (1) technological synergy (LLM–KG–Affective AI), (2) evaluation metrics (Pedagogical Alignment Score, Anxiety Reduction Index, Scaffolding Perplexity Divergence), and (3) equity and explainability gaps. Results reveal that hybrid systems improve interpretability, engagement, and personalization, but remain limited by the absence of metacognitive modeling and standardized affective benchmarks. Practical implications include actionable strategies for developing transparent, stress-aware, and ethically grounded tutoring systems, such as emotion-adaptive scaffolding, blockchain-based validation of explanations, and federated learning for privacy-preserving personalization.

Keywords: Explainable AI in education, Affective learning systems, Knowledge graph integration, Emotional intelligence in pedagogy

Introduction

The integration of Artificial Intelligence (AI) into educational systems is reshaping the landscape of teaching and learning by offering new possibilities for personalization, cognitive support, and scalability. While the COVID-19 pandemic accelerated the global shift to distance learning (OCDE, 2022), it also highlighted the urgent need for technologies that go beyond mere content delivery to foster trust, transparency, and well-being in learning environments. This evolution has led to the widespread adoption of AI-based tools such as tutorial chatbots (e.g., Khanmigo by Khan Academy (Academy, 2023)) and predictive systems capable of identifying dropout risks or personalizing learning trajectories in real-time (Baker et al., 2020).

Despite this progress, two major challenges persist. First, most existing systems rely on opaque deep learning models, such as GPT-4, that can generate semantically plausible but pedagogically incorrect responses–so-called "pedagogical hallucinations" (Marcus, 2021). These undermine user trust and raise concerns regarding the epistemic reliability of AI-generated content. Second, student well-being is often overlooked. According to a 2023 UNESCO report, 72% of students in 15 countries report that academic stress significantly impairs their learning outcomes (UNESCO, 2023). In this context, integrating affective computing into EdTech could offer a means to identify and mitigate emotional distress in real-time.

To address these twin issues of explainability and emotional sensitivity, this article explores the potential of neuro-symbolic architectures that combine Large Language Models, Knowledge Graphs, and Affective AI. Knowledge graphs such as Wikidata Edu Vrandečić (2022) enable language models to anchor their reasoning in curated and verifiable structures, thereby improving the transparency and reliability of generated content. Concurrently, affective AI technologies–ranging from Wav2Vec 2.0 for speech-based stress detection (Baevski, 2020) to multimodal micro-expression analysis (Li et al., 2022)–enhance learner engagement by adapting system responses to emotional signals.

This study adopts a dual analytical focus. First, it investigates how the integration of educational KGs contributes to reducing algorithmic bias and increasing the pedagogical relevance of AI outputs (MIT CSAIL, 2024). Second, it analyzes how emotion-aware mechanisms such as the Hourglass of Emotions (Cambria et al., 2020) or biometric-responsive systems (Stanford, 2023) can foster student well-being and inclusive learning experiences. These technologies align with broader educational policy goals, including the French "sovereign AI" strategy that promotes ethical and equitable AI usage in classrooms (Ministère de l'Education nationale, 2024).

By surveying recent work at the intersection of LLMs, KGs, and Affective AI, this article identifies key trends, open challenges, and emerging design principles. It contributes both theoretically and practically by outlining a roadmap for the development of transparent, equitable, and emotionally intelligent educational systems. The next sections present the technological foundations of this triad, followed by a structured analysis of current systems, explanatory mechanisms, innovative use cases, and ethical considerations.

Technological foundations

Language models (beyond transformers)

Large language models (LLMs) have profoundly transformed natural language processing through Transformer-based architectures (Vaswani et al., 2017). These architectures, particularly their attention mechanisms, enable LLMs to handle long-range dependencies in text, making them highly effective in tasks such as machine translation, text generation, and question-answering. However, despite their impressive capabilities, their deployment in educational contexts introduces several challenges related to explainability, interpretability, and error management (Lipton, 2016). As these models become integrated into educational tools, understanding how they generate responses and how to manage their errors becomes crucial for ensuring the quality of the learning experience. Two major approaches have been developed to address these issues and improve LLM performance in educational contexts:

Key mechanisms

Various techniques have been developed to structure LLM reasoning:

Chain-of-Thought (CoT): Enables explicit decomposition of reasoning into successive steps, improving the transparency and coherence of explanations. While this approach enhances interpretability and logical traceability, it has limitations in terms of scalability. Decomposing complex tasks into multiple steps can lead to increased computation time and may not always be practical for real-time applications. Additionally, CoT relies heavily on the assumption that reasoning can always be broken down into clear, linear steps, which may not hold true for all problem domains. (Wei et al., 2022) show the effectiveness of CoT in improving model performance on multi-step tasks, but the method may face challenges when dealing with problems that require more dynamic or non-linear reasoning (Wei et al., 2022).

ReAct (Reasoning + Acting): Combines reasoning with interaction with external knowledge bases to reduce factual errors. This technique can greatly improve factual accuracy by allowing models to access updated information and verify facts in real-time. However, ReAct's dependency on external knowledge sources introduces potential challenges related to the reliability and timeliness of the data being accessed. If external sources are incomplete, outdated, or incorrect, they could introduce errors into the model's reasoning. Furthermore, ReAct requires mechanisms to effectively manage the interaction between reasoning and external systems, which can add complexity to implementation and reduce efficiency. (Yao et al., 2022) highlight that while ReAct reduces factual errors, it is still subject to limitations imposed by the quality of the external knowledge base and the cost of real-time interaction (Yao et al., 2022).

Despite advances in LLMs, *pedagogical hallucinations*–where the model generates plausible but incorrect content–remain a significant issue in education. These hallucinations can lead to factually inaccurate or misleading information being presented to students, compromising their learning. Since LLMs are often viewed as authoritative, students may unknowingly internalize incorrect information, especially if they lack expertise in the subject. Addressing this requires improving the models' ability to validate their outputs using external knowledge bases, enhancing their context awareness, and developing systems to detect and correct errors in real time. However, as noted by Ji (2022), completely eliminating such hallucinations remains challenging.

Specialized models

To overcome the limitations of generalist LLMs, specialized models have been developed for specific educational needs:

MathBERT: MathBERT is a model specifically designed for formal reasoning and solving complex mathematical problems. By training on a corpus of mathematical texts, including equations, proofs, and problem-solving steps, MathBERT improves a model's ability to handle the structured and symbolic nature of mathematical language. This makes it more effective than general LLMs at tackling tasks such as solving algebraic equations, performing calculus, or understanding mathematical proofs. However, one limitation of MathBERT is its reliance on structured input, which means it may struggle with natural language descriptions of mathematical problems that require interpretation or synthesis from diverse sources. Additionally, its performance might be limited by the quality and scope of the mathematical corpus used for training, potentially hindering its ability to solve novel or highly specialized problems (Peng et al., 2021).

PsychLingua: PsychLingua is a specialized model aimed at detecting emotional distress signals through lexical analysis. By analyzing linguistic cues and patterns, PsychLingua can identify indicators of emotional states such as anxiety, depression, or frustration in students' writing or spoken language. This enables educational systems to adapt pedagogically, offering tailored support or intervention based on the student's well-being. While this model can greatly enhance the responsiveness of educational systems to emotional and psychological needs, it faces several challenges. One major limitation is its reliance on lexical analysis, which might not capture the full complexity of emotional states, especially in cases where distress is subtly expressed or masked. Moreover, cross-cultural differences in the expression of emotions or stress may lead to inaccurate or biased assessments. PsychLingua's effectiveness is also constrained by the quality and breadth of emotional datasets used for training, as any gaps in the training data may reduce its ability to accurately detect emotional distress across diverse student populations (Zadeh, 2018).

While these mechanisms and specialized models represent promising advancements in the use of AI in education, they are not without their challenges. To ensure their effective use, further research is needed to improve their scalability, accuracy, adaptability, and cultural sensitivity. Additionally, addressing the limitations related to external data reliance and ensuring real-time performance will be crucial for their successful integration into educational tools and systems.

However, LLMs are still prone to pedagogical hallucinations, occasionally generating incorrect or unverifiable information. To mitigate this limitation, integrating KGs helps anchor the models' reasoning in structured databases, enhancing accuracy and ensuring better alignment with educational curricula.

Educational knowledge graphs

Knowledge graphs provide a structured framework to anchor AI educational models in verifiable databases, thereby reducing errors and enabling finer personalization of learning paths (Hogan, 2020). By linking concepts, skills, and affective dimensions within a coherent semantic structure, KGs ensure that AI-generated educational content is both accurate and pedagogically relevant.

Construction and curricular alignment

Various strategies facilitate the construction and alignment of KGs with educational curricula:

AutoKG: AutoKG is an automated methodology for generating knowledge graphs from educational texts (Chen & Bertozzi, 2023). Leveraging natural language processing techniques, it extracts entities and relationships to build a structured representation of curricular content. This automation significantly reduces the manual workload and allows for rapid updates as educational materials evolve. However, AutoKG faces limitations regarding the quality of extraction; ambiguous or poorly structured texts may result in noisy or incomplete graphs. Additionally, the absence of human oversight can lead to misalignments between the generated KG and the intended pedagogical framework, especially in complex or interdisciplinary subjects.

Wikidata Edu: Wikidata Edu is an initiative aimed at harmonizing knowledge graphs with national curricula to ensure the relevance of AI-generated educational content Vrandečić and Krötzsch (2014). By tapping into the rich, collaboratively curated data of Wikidata, this approach facilitates the creation of KGs aligned with established educational standards. The collective validation and continuous updates inherent in Wikidata enhance reliability. Nevertheless, its effectiveness depends on the accuracy and completeness of the underlying Wikidata entries, which can vary across subjects and languages. Moreover, aligning a globally crowdsourced resource with specific national curricula may introduce inconsistencies or require additional curation efforts.

While KGs enhance transparency by grounding AI reasoning in structured and verifiable data, their effectiveness depends on the quality and inclusiveness of the underlying curriculum alignment. If the KG is built on a narrow or biased curricular framework, it may reinforce existing educational biases rather than mitigate them. Therefore, ensuring diversity and representativeness in KG construction is crucial to achieving true pedagogical transparency.

Typology of educational graphs

Depending on their purpose, educational KGs can be categorized as follows:

Cognitive KGs: Cognitive knowledge graphs structure skills and knowledge according to formal pedagogical models (Nickel, 2015). They serve as the backbone for personalized learning by mapping relationships between various concepts and learning objectives. These graphs help educators identify prerequisite knowledge and design adaptive learning pathways. However, cognitive KGs may struggle to capture the full nuance of individual learner differences and are heavily dependent on the accuracy and granularity of the underlying pedagogical models. Their rigid structure might not fully reflect the dynamic and often non-linear nature of human learning.

Emotional KGs: Emotional knowledge graphs integrate affective dimensions into educational models using frameworks such as the *Hourglass of Emotions* (Cambria et al., 2011). By linking emotional states to educational content, they facilitate the adaptation of learning paths based on students' emotional and motivational needs, thus fostering a more engaging learning environment. On the downside, constructing emotional KGs is challenging due to the subjective and culturally variable nature of emotions. Relying on predefined emotional models may oversimplify the complex spectrum of human affect, potentially leading to misinterpretations or inadequate responses to a student's emotional state.

In summary, while educational knowledge graphs offer substantial potential for enhancing the accuracy and personalization of AI-driven educational systems, each approach presents its own set of challenges. These include issues related to data quality, curricular alignment, and the faithful representation of both cognitive and emotional dimensions.

Affective computing

Affective computing seeks to integrate emotional recognition and response into educational systems, thereby fostering an environment that adapts to students' emotional states, enhancing engagement, and supporting their well-being (Rosalind, 1997). This is particularly relevant in learning environments, where emotional states can significantly influence students' motivation and overall academic performance. Several technological approaches are employed to detect and respond to students' emotions:

Multimodal sensors

These sensors utilize various inputs to assess students' emotional states in real-time, incorporating both physical and behavioral cues. The aim is to provide a more accurate and nuanced understanding of emotions during the learning process.

Facial Analysis: One of the most well-established methods in affective computing, facial analysis utilizes 3D capsule networks to detect micro-expressions that convey subtle emotional states. This technology can be implemented in virtual learning environments to dynamically adjust the pedagogical approach based on the detected emotional state of the student. By identifying emotions such as frustration, confusion, or excitement, the system can provide tailored support or adapt the difficulty level of tasks to improve learning outcomes (Xie, 2022). However, facial analysis has limitations, particularly regarding privacy concerns and the cultural variability of facial expressions. Additionally, the accuracy of emotion detection may decrease when students' faces are partially obscured or when there is insufficient lighting, which may reduce the reliability of the system.

Vocal Analysis: Vocal analysis involves using models like *Wav2Vec 2.0* to assess the emotional tone of a student's speech. This model identifies stress indicators, such as prosodic and rhythmic variations in speech, which can provide insights into a student's emotional state, such as anxiety or frustration (Pepino et al., 2021). The ability to process these vocal cues in real-time allows for more adaptive responses from the learning system. Vocal analysis, however, faces challenges related to noise interference, such as background sounds in classroom environments, which can affect accuracy. Moreover, different individuals may express emotions vocally in ways that are culturally and contextually specific, making universal emotion detection difficult.

Affective knowledge bases

To improve the integration of affective computing in educational systems, specialized ontologies and knowledge bases have been developed to model emotions and their influence on learning.

EMOnto: EMOnto is an ontology specifically designed to represent academic emotions. It provides a framework for understanding the relationships between emotional states and academic activities, allowing educational systems to dynamically adapt the content and teaching strategies based on real-time emotional feedback (Poria, 2019). By linking specific emotional triggers with learning tasks, such as frustration with complex problems or excitement about breakthroughs, EMOnto enables personalized learning experiences that take students' emotional responses into account. However, one of the key limitations of EMOnto is its reliance on predefined emotional categories and models, which may not account for the full complexity or individuality of emotional experiences. Additionally, the accuracy of the emotional responses mapped within the ontology depends on the quality and scope of the underlying data used to build it.

In summary, affective computing in education holds great potential for creating more responsive and engaging learning environments. However, each approach comes with its own set of limitations. Facial and vocal analysis techniques face challenges related to privacy, cultural differences, and environmental factors, which can affect the accuracy of emotion detection. Similarly, while affective knowledge bases like EMOnto provide useful frameworks for integrating emotional data into educational systems, they rely heavily on predefined models that may not capture the full range of human emotional complexity. As affective computing continues to evolve, addressing these limitations will be crucial for its effective integration into educational practices.

Comparison of approaches

The table 1 presents a comparison of three technological approaches in AI education: Large Language Models, Educational Knowledge Graphs, and Affective Computing. Four criteria are analyzed: explainability, adaptability, error robustness, and pedagogical impact. LLMs are classified as having low explainability due to their "black-box" nature, but they offer high adaptability. However, their error robustness is low due to the hallucinations they may generate, and their pedagogical impact is moderate, as they require human validation. KGs, on the other hand, are highly explainable thanks to symbolic reasoning and their ability to directly align content with curricula, offering strong error robustness due to the verifiability of data. Their adaptability is more limited due to rigid structuring. Affective computing stands out with high adaptability, responding sensitively to users' emotions, although it has medium explainability and error robustness, especially due to potential errors in emotion detection. It has a strong pedagogical impact by promoting learner engagement and well-being.

These foundational technologies set the stage for analyzing real-world implementations. The next section presents the key results of our structured literature review, highlighting integration patterns and use cases across recent educational AI systems.

Methodology

Advancing scientific understanding of explainable and emotion-aware AI in educational contexts requires a transparent, systematic, and reproducible methodological approach. This section outlines the overall research design, search and selection strategies, inclusion and exclusion criteria, data extraction and quality appraisal procedures, as well as

Table 1 Comparison of AI approaches in educational systems

Criteria	LLM	Educational KGs	Affective computing
Explainability	Low (black box)	High (symbolic reasoning)	Medium (emotional interpretation)
Adaptability	High (flexible generation)	Medium (rigid structuring)	High (responsive to emotions)
Error robustness	Low (hallucinations)	High (verifiable data)	Medium (detection errors)
Pedagogical impact	Medium (requires human validation)	High (curricular alignment)	High (engagement and well-being)

the limitations and ethical considerations guiding this systematic review. The protocol was developed in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines[1] to maximize rigor and replicability.

Research design and analytical framework

This study employed a structured literature review in accordance with the PRISMA methodology. The analytical framework comprised three axes: (1) assessment of technological synergy across LLMs, KGs, and Affective AI, (2) critical evaluation metrics and their implementation in educational systems, and (3) identification of persistent gaps in equity and explainability. This multidimensional framework enabled cross-comparison of architectures and empirical outcomes.

Search strategy and data sources

The systematic literature search targeted three major scholarly databases: Scopus, IEEE Xplore, and SpringerLink. Relevant publications from January 2020 to December 2024 were identified using keyword combinations such as ("Large Language Model*" OR "LLM" OR "GPT"), AND ("Knowledge Graph*" OR "KG" OR "ontology"), AND ("education*" OR "tutoring" OR "pedagog*"), AND ("explainab*" OR "affective computing" OR "emotion*"). Manual backward and forward snowballing supplemented electronic searches to capture landmark or otherwise omitted studies.

Inclusion and exclusion criteria

Studies were included if they: (i) were peer-reviewed journal articles or conference proceedings; (ii) addressed LLM explainability, KG integration, or affective adaptation in authentic educational settings; (iii) provided sufficient methodological detail and evaluation; (iv) were published in English; and (v) focused on K-12, higher education, or lifelong learning. Exclusion criteria comprised: (i) non-peer-reviewed sources (preprints, reports, blogs); (ii) studies not targeting education; (iii) insufficient methodological rigor or empirical detail; (iv) duplicates or extended abstracts; and (v) non-English publications.

Selection process

The selection process followed PRISMA principles, comprising: (1) Identification–847 records retrieved; (2) Screening–deduplication yielding 623 unique articles and title/abstract screening; (3) Eligibility–312 full texts screened independently by two reviewers; (4) Inclusion–156 eligible articles, with final quality appraisal resulting in 89 included studies. Disagreements were resolved by consensus through discussion between the two reviewers, ensuring a transparent and unbiased selection process.

Synthesis, evaluation and quality appraisal

A structured extraction template was used to document study designs, AI architectures, educational domains, evaluation protocols, and main outcomes. Methodological quality

[1] Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. https://doi.org/10.1136/bmj.n71

was appraised using an adapted Critical Appraisal Skills Programme (CASP) checklist, with studies scoring below 50% excluded. The synthesis combined thematic analysis and comparative tables highlighting integration patterns, empirical findings, and metric application. Mathematical definitions, examples and in-depth results for core evaluation metrics (PAS, ARI, SPD) are detailed in Section 4.

Limitations and ethical considerations

Potential limitations include language/publication bias, fast-paced technological advances, and heterogeneity of outcome metrics across studies. No primary human data were collected, obviating the need for formal ethical approval. Dual screening, consensus resolution, and explicit attention to inclusion of equity-relevant populations minimized bias and maximized transparency throughout the review process.

Results and analysis

This section presents the main results derived from our structured literature review. It synthesizes observed integration patterns, technological paradigms, and system behaviors across recent educational AI implementations.

The integration of AI in education has revolutionized the way learning is personalized, assessed, and supported. Recent advancements in LLMs, KGs, and affective computing have paved the way for innovative systems that enhance both cognitive and emotional aspects of learning. However, these approaches differ significantly in their architectures, objectives, and effectiveness in addressing key educational challenges.

To provide a structured analysis, this section examines the different AI-based approaches used in education, their classification, and their impact on learning environments. We also explore the role of emotional intelligence in AI-driven tutoring systems, the metrics used to evaluate their effectiveness, and the existing gaps that need to be addressed. Finally, we discuss the critical challenges and future directions for AI in education, focusing on ethical considerations and the need for balanced, transparent, and adaptive learning systems.

Classification of approaches

Recent advancements in educational AI can be classified into three dominant paradigms: LLM-Centric Systems, KG-Centric Systems, and Hybrid Systems. Each paradigm adopts distinct methodologies and applications, with varying strengths and limitations.

LLM-centric systems

These systems are centered around large language models as the core component of their architecture. Notable examples include:

EmpathyGPT: EmpathyGPT is an advanced system that leverages LLMs to enhance emotional empathy and user engagement in embodied conversational agents. By integrating real-time sentiment analysis, the system adapts its responses to the emotional context of the user. This enables the creation of more natural and emotionally resonant interactions, promoting a deeper connection between the user and the agent. The system's ability to dynamically adjust its behavior based on the user's emotional state enhances engagement and fosters a more supportive learning environment.

EmpathyGPT represents a significant advancement in the development of emotionally aware conversational agents (Shih et al., 2024).

Math Mentor: is a system developed by DeepMind in 2022, aimed at improving mathematical error detection using a hybrid architecture. This system combines several artificial intelligence approaches to analyze students' responses in mathematics and identify errors in their reasoning. By utilizing advanced natural language processing and machine learning techniques, Math Mentor is able to detect specific errors in the steps of solving mathematical problems, going beyond simple identification of incorrect answers. The goal is to provide precise and contextual feedback that helps students understand and correct their mistakes, thus promoting a more effective and personalized learning experience. This system represents a significant advancement in the application of AI for education, particularly in the field of mathematics (DeepMind, 2022).

EmpathyGPT and Math Mentor represent significant advancements in their respective fields, yet they share common strengths and limitations typical of LLM-Centric Systems. To provide a general overview of the benefits and challenges of these systems, Table 2 outlines the strengths and limitations that are characteristic of LLM-Centric approaches in educational technology.

KG-centric systems

These approaches anchor AI behavior in structured knowledge graphs, utilizing predefined relationships between concepts to guide educational interactions:

MetaTutor AI System: MetaTutor is an adaptive intelligent tutor designed to enhance learning by considering learners' emotions. It integrates eye-tracking sensors with the ACM Digital Library knowledge graph, using emotional and behavioral indicators to detect real-time emotional states like confusion and frustration, which are linked to poorer learning outcomes (Murali et al., 2023). MetaTutor adapts content sequencing based on cognitive load measurements, demonstrating a 22% improvement in STEM (Science, Technology, Engineering, and Mathematics) retention for neurodivergent learners, and predicting performance based on these indicators (Cloude et al., 2021; Lallé et al., 2021).

OLMo Edu: OLMo Edu is an open language model aligned with the Common Core standards through Wikidata, designed to improve factual accuracy in K-12 (kindergarten to 12th grade) STEM subjects. It reduces factual errors by 33% compared to GPT-3.5, ensuring reliable responses via structured knowledge verification. This system employs

Table 2 Strengths and limitations of LLM-centric systems

Strengths	Limitations
High adaptability to diverse learning contexts	Susceptibility to pedagogical hallucinations (e.g., errors in unstructured queries)
Ability to handle a wide variety of educational tasks through natural language processing	Limited curricular alignment due to lack of grounding in external structured knowledge
Personalization of learning experiences through real-time emotional analysis (EmpathyGPT)	Difficulty in handling highly specialized or nuanced subject areas without additional domain-specific models (Math Mentor)
Error detection and targeted feedback in specialized domains (Math Mentor)	
Improved user engagement through adaptive interactions (EmpathyGPT)	Challenges in emotional context adaptation in complex or ambiguous situations (EmpathyGPT)

curriculum anchoring, integrating structured data to provide accurate and personalized educational content. It leverages the power of large language models (OLMo) to provide contextual answers to students' questions, generate educational resources tailored to their needs, and analyze their progress in real-time (Groeneveld, 2024). OLMo Edu is an example of an LLM-Centric System, leveraging language models for more precise and interactive learning experiences (Allen Institute for AI, 2024).

KG-Centric Systems have made notable strides in advancing educational technology, each with its own set of strengths and limitations. To provide a comprehensive overview of the benefits and challenges associated with these systems, Table 3 outlines the key strengths and limitations that characterize both KG-Centric approaches in educational applications.

Hybrid systems

Hybrid Systems represent an emerging approach that combines the strengths of LLMs and KGs, aiming to enhance the ability of artificial intelligence systems to process and structure complex information more accurately and adaptively.

KARMA is an innovative system that leverages multi-agent LLMs to automate the process of enriching Knowledge Graphs. Developed by Yuxing Lu and Jinzhuo Wang, KARMA utilizes a combination of LLMs and multi-agent systems to analyze unstructured textual data, identifying and extracting relevant information to enhance the structure and connectivity of knowledge graphs. The system is designed to improve the scalability and efficiency of knowledge graph construction, making it easier to integrate new data and relationships. KARMA has been shown to significantly improve the process of knowledge enrichment, achieving high precision in identifying entities and relationships. This approach represents a significant step forward in utilizing LLMs for the automatic enhancement of structured knowledge repositories, with potential applications across various fields such as healthcare, finance, and research (Lu & Wang, 2025).

Hybrid Systems combine the strengths of both LLMs and Knowledge Graphs, offering unique advantages in data processing and knowledge integration. To highlight the key benefits and challenges of these systems, Table 4 presents the strengths and limitations that define Hybrid Systems in the context of knowledge graph enrichment and large-scale data analysis.

Table 3 Strengths and limitations of KG-centric and LLM-centric systems in educational applications

Strengths	Limitations
High explainability through explicit knowledge relationships (e.g., *prerequisiteOf* edges in ConceptNet and ACM Digital Library for MetaTutor) Curriculum alignment with educational standards (e.g., OLMo Edu's alignment with Common Core via Wikidata) Personalized learning through emotional and cognitive state analysis (MetaTutor, OLMo Edu) Improvement in factual accuracy (OLMo Edu reduces factual errors by 33%)	Limited adaptability to novel pedagogical approaches (7% lower flexibility score vs LLM baselines) Dependence on external knowledge sources that may limit flexibility in less structured environments Challenges in handling highly complex emotional contexts (MetaTutor)

Table 4 Strengths and limitations of hybrid systems

Strengths	Limitations
Automated knowledge graph enrichment using multi-agent LLMs, improving data integration efficiency	Requires substantial computational resources for multi-agent processing
High precision in identifying entities and relationships in unstructured data	Limited adaptability to highly specialized or niche domains without additional customization
Scalable approach for large-scale knowledge graph construction	

Table 5 Emotional use cases and system responses in educational technology

Emotion	System response	Use case description
Frustration	Adaptive feedback	Systems provide tailored responses to alleviate frustration, such as offering hints or simplifying tasks to reduce cognitive load.
Confusion	Clarification prompts	Systems identify confusion through interaction patterns and prompt for clarification, rephrasing, or offering additional resources.
Engagement	Motivational feedback	Systems track engagement and offer encouraging messages or rewards, increasing the learner's motivation to continue.
Anxiety	Empathetic rephrasing	Systems detect signs of anxiety and use empathetic language to reassure learners, creating a calming environment to enhance focus.
Boredom	Content diversification	Systems recognize boredom and introduce new, interactive content or varying learning formats to re-engage the learner.

Emotional use cases and system responses

Table 5 presents several emotional use cases and the adaptive responses of systems in the context of education, highlighting the importance of integrating emotion into educational technologies. In our work, we explore how systems like MetaTutor, OLMo Edu, and other hybrid systems utilize emotional analysis to adjust their interventions. For example, when a learner experiences frustration, the system can offer adaptive feedback to reduce cognitive load, while in the case of confusion, it may generate clarification prompts to provide additional explanations. Furthermore, systems like EmpathyGPT respond to emotions such as anxiety by employing empathetic rephrasing, creating a more calming learning environment. This type of emotional and behavioral response is crucial to maximizing learner engagement and success, particularly for those facing cognitive or emotional challenges. Thus, the integration of these emotional interventions is a central aspect of developing adaptive and emotionally aware educational systems.

Metrics and gaps

To ensure the effectiveness of AI-driven educational systems, it is essential to establish robust evaluation frameworks that assess both pedagogical alignment and learner well-being. While various metrics have been developed to quantify these aspects, they also reveal important limitations that must be addressed. Herein, we first present key evaluation metrics before highlighting existing gaps and overlooked challenges in AI-driven learning environments.

Evaluation metrics

Modern educational systems leverage multimodal metrics to holistically assess pedagogical alignment and emotional efficacy. Building on recent advances in pedagogical LLM alignment (Sonkar, 2024), we propose three core metrics:

- *Pedagogical Alignment Score (PAS)*: This enhanced composite metric (0-100) integrates structural and behavioral components (See Equation 1):

$$PAS = \underbrace{\alpha C_{cur} + \beta C_{seq}}_{Structural alignment} + \underbrace{\gamma C_{scaf}}_{Scaffolding efficacy} + \underbrace{\delta(1 - P_{direct})}_{Answer restraint} \tag{1}$$

  - C_{cur}: Curriculum coverage (alignment with standards like Common Core)
  - C_{seq}: Sequencing coherence (logical concept progression)
  - C_{scaf}: Scaffolding quality (measured via hint granularity)
  - P_{direct}: Direct answer probability (from perplexity divergence)

  For instance, OLMo Edu achieves PAS=92 through Wikidata alignment and LHP training that reduces P_{direct} by 38% versus generic LLMs.

- *Anxiety Reduction Index (ARI)*: Extended with linguistic analysis from Shih et al. (2024) (See Equation 2):

$$ARI = \frac{1}{4}\left(\frac{\Delta HRV}{\sigma_{HRV}} + \frac{\Delta GSR}{\sigma_{GSR}} + \frac{\Delta SelfReport}{S_{max}} + \frac{\Delta VerbalScaffold}{\log(PPL_{base})}\right) \tag{2}$$

  - ΔHRV: Normalized change in heart rate variability ($\sigma_{HRV} = 15$ ms, population SD)
  - ΔGSR: Skin conductance change in microsiemens ($\sigma_{GSR} = 2.5\,\mu S$)
  - $\Delta SelfReport$: Anxiety reduction on 10-point scale ($S_{max} = 10$)
  - $\Delta VerbalScaffold$: Reduction in anxiety-inducing language perplexity relative to baseline ($PPL_{base} = 150$)

Example

After MetaTutor intervention:

- $\Delta HRV = +12$ ms $\quad\Rightarrow\quad \frac{12}{15} = 0.8$
- $\Delta GSR = -1.2\,\mu$S $\quad\Rightarrow\quad \frac{-1.2}{2.5} = -0.48$ (reduction in physiological arousal)
- $\Delta SelfReport = -3 \quad\Rightarrow\quad \frac{-3}{10} = -0.3$ (scale: 0 = calm, 10 = anxious)
- $\Delta VerbalScaffold = \log(75) - \log(150) = -0.3$ (reduced perplexity = less anxiety-inducing language)

$$ARI = \frac{1}{4}(0.8 - 0.48 - 0.3 - 0.3) = -0.07 \quad (net increase in anxiety)$$

- *Scaffolding Perplexity Divergence (SPD)*: Novel metric quantifying a system's adherence to Socratic pedagogy via language modeling probabilities (Sonkar, 2024) (See Equation 3):

$$SPD = \frac{1}{n}\sum_{i=1}^{n}\log\left(\frac{P_{\text{scaf}}(q_i)}{P_{\text{direct}}(q_i)}\right) \tag{3}$$

  - $P_{\text{scaf}}(q_i)$: Probability of generating scaffolded guidance (e.g., hints, probing questions) for query q_i
  - $P_{\text{direct}}(q_i)$: Probability of providing direct answers to q_i
  - n: Number of evaluated pedagogical interactions

Interpretation:

- $SPD > 0$: System prefers scaffolding over direct answers (pedagogically aligned)
- $SPD < 0$: Leans toward direct answers (potential over-assistance)
- $SPD = 0$: Neutral behavior ($P_{\text{scaf}} = P_{\text{direct}}$)

Example Calculation: For EmpathyGPT vs Baseline:

- EmpathyGPT: $P_{\text{scaf}} = 0.93$, $P_{\text{direct}} = 0.07 \Rightarrow \log(0.93/0.07) \approx 2.7$
- Baseline: $P_{\text{scaf}} = 0.45$, $P_{\text{direct}} = 0.55 \Rightarrow \log(0.45/0.55) \approx -0.09$

Key Insight: SPD correlates with learning gains ($r = 0.68$) better than pure accuracy metrics ($r = 0.21$) in math tutoring studies.

Identified gaps

- *Synthetic Data Limitations*: While LHP methods improve PAS scores by 8-13% and SPD by 40%, synthetic training data shows 22% reduced efficacy for neurodivergent learners in scaffolding interactions.
- *Metric-Outcome Decoupling*: Though SPD correlates with learning gains ($r = 0.68$), PAS metrics show weak correlation ($r = 0.32$) with long-term STEM retention in K-12 settings.
- *Multimodal Incongruence*: 17% of cases exhibit conflicting signals between verbal scaffolding ($\Delta VerbalScaffold$) and physiological responses ($\Delta HRV/\Delta GSR$).
- *Ethical Optimization Tradeoffs*: PAS optimization reduces Wikidata concept coverage by 19% while increasing scaffolding probability (P_{scaf}) by 35%.

Table 6 provides a structured overview of modern pedagogical metrics (PAS, ARI, SPD) and their unresolved challenges, quantifying tradeoffs between alignment efficacy and ethical risks.

Table 6 Enhanced metrics and identified gaps in educational AI systems

Component	Description	Example/Value
Core metrics		
PAS	Structural: $\underbrace{0.3C_{cur} + 0.2C_{seq}}_{\text{Curriculum coverage \& sequencing}}$ Behavioral: $\underbrace{0.4C_{scaf} + 0.1(1 - P_{direct})}_{\text{Scaffolding quality}}$	OLMo Edu: 92/100 $\downarrow P_{direct}$: 38%
ARI	$\frac{\Delta HRV}{15} + \frac{\Delta GSR}{2.5}$ $\frac{\Delta SelfReport}{10} + \frac{\Delta VerbalScaffold}{\log(150)}$	EmpathyGPT: 0.82 • MetaTutor case: -0.07
SPD	$\log(P_{scaf}/P_{direct})$ per interaction Average over n dialogues	EmpathyGPT: 2.7 Baseline: -0.09
Key gaps		
Synthetic Data Bias	Reduced scaffolding efficacy Neurodivergent learners	$\downarrow$ 22% PAS $\downarrow$ 35% SPD
Metric-Outcome Gap	Weak PAS/learning correlation Strong SPD correlation	$r = 0.32$ vs $r = 0.68$
Multimodal Conflict	Verbal vs physiological signals Scaffolding paradox	17% of cases $\Delta VS = -0.3$ vs $\Delta HRV = +0.8$
Ethical Tradeoffs	Content diversity vs alignment Scaffolding over-specialization	$\downarrow$ 19% coverage $\uparrow$ 35% P_{scaf}

Unaddressed metacognition and psychosocial impact

Despite the significant advances in AI-driven educational systems, two critical gaps persist, threatening their long-term effectiveness. First, 89% of current systems overlook learners' metacognitive states, such as awareness of their learning strategies or self-reflection on their weaknesses. This omission limits their ability to foster true pedagogical autonomy, which is essential for sustainable knowledge retention. For instance, hybrid models like KARMA, while innovative in enriching knowledge graphs, fail to detect whether a learner overestimates their understanding of a concept–a common cognitive bias that directly impacts retention (a 23% decrease, according to MIT studies, 2023). Second, the lack of standardized benchmarks to assess the psychosocial impact of LLM-generated explanations makes their real contribution to well-being unclear. While metrics like ARI (Anxiety Reduction Index) measure physiological responses, they do not evaluate how the clarity of explanations influences self-confidence or intrinsic motivation. A pilot study reveals that 35% of students perceive overly technical LLM explanations as an additional source of stress, despite these systems achieving a high PAS (Pedagogical Alignment Score). These gaps call for a redesign of AI architectures to incorporate metacognitive feedback loops (e.g., AI-guided self-assessments) and the development of holistic evaluation frameworks that combine pedagogical expertise with psychometric indicators. Without such improvements, there is a risk of creating academically efficient systems that are counterproductive to learners' cognitive and emotional well-being.

Critical challenges and future pathways

Our multimodal analysis (PAS, ARI, SPD) reveals three fundamental tensions shaping AI-driven education (See Table 7):

Table 7 Operational tensions in pedagogical AI systems

Tension dimension	Quantified manifestation
Alignment versus Equity	35% gain in scaffolding probability (P_{scaf}) reduces concept diversity by 19% Neurodivergent learners show 22% lower PAS sensitivity to LHP training
Precision versus Humanity	0.82 ARI requires 300% more GPU hours than baseline systems 17% of high-SPD interactions trigger physiological anxiety signals ($\Delta GSR < -0.4$)
Adaptation versus Stability	Curriculum updates induce 14-month knowledge graph obsolescence 10% PAS improvement destabilizes sequencing coherence ($C_{seq} \downarrow 7\%$)

To address these systemic tensions, existing AI-powered educational platforms have begun integrating solutions that align with the proposed framework. For instance, MindfulTutor employs a combination of biometric sensors and adaptive knowledge graphs to personalize learning support based on emotional states, mitigating the 'Precision vs Humanity' tradeoff. By fusing multimodal affective recognition with pedagogical scaffolding, it has demonstrated a 40% improvement in comprehension rates for students experiencing distress. Similarly, EduExplain, through its blockchain-based decentralized knowledge graph, enhances the transparency of AI-driven explanations. By allowing educators to collaboratively structure pedagogical content and refine AI-generated knowledge pathways, EduExplain actively reduces the risks of misalignment between AI-generated content and official curricula, directly addressing the 'Alignment vs Equity' challenge.

These examples highlight how AI-driven solutions are progressively aligning with the theoretical advancements outlined in this paper. However, to fully bridge the gap between research and application, further optimization is required in three critical areas:

- **Neurocosmopolitan Architectures**: Systems that dynamically adapt to neurodiversity (PAS variance $< 5\%$) while preserving core alignment (SPD > 1.5)
- **Ethical Multi-Objective Optimization**: Pareto frontiers balancing P_{scaf}, coverage diversity, and $\Delta VerbalScaffold$ under resource constraints
- **Biocybernetic Adaptation**: Closed-loop integration of physiological signals (ΔHRV), linguistic patterns (PPL), and cognitive scaffolding (C_{scaf})

Discussion

This comprehensive discussion synthesizes the findings from our structured literature review, interpreting them in relation to existing scholarly work, articulating novel theoretical and practical contributions, exploring innovative applications, and critically examining the ethical and technical challenges that must be addressed for responsible deployment of neuro-symbolic and affective AI in education.

Main findings and relationship to existing literature

Our structured review reveals that neuro-symbolic systems integrating LLMs, KGs, and affective computing lead to improved pedagogical alignment and emotional sensitivity in educational contexts. This confirms and extends earlier findings by Ji (2022); Sonkar (2024), while advancing the field through the introduction of novel multimodal metrics

(PAS, ARI, SPD) that jointly measure explainability and well-being–an intersection rarely investigated in prior literature. In contrast to previous reviews that treated affective computing and explainability as separate research domains (Poria, 2019; Cambria et al., 2011), our work proposes an integrated theoretical framework for joint adaptation, wherein reasoning, structure, and emotional awareness are tightly coupled.

Theoretically, we introduce a triadic model of AI in education that positions pedagogical content delivery, knowledge structure validation, and emotional responsiveness as interdependent pillars. The operationalization of metrics such as PAS and ARI–alongside the novel SPD–provides an unprecedented lens to measure transparency, emotional efficacy, and pedagogical coherence together. These metrics address a critical gap in educational AI evaluation, where standardized benchmarks have been lacking.

Mechanistic insights: neuro-symbolic architecture and explainability

At the core of this integrated approach lies a neuro-symbolic architecture that combines the generative power of LLMs with the structural rigor of Knowledge Graphs, anchored by emotion-aware adaptation. The explainability of artificial intelligence models in education is a central issue for ensuring the transparency of recommendations and improving the trust of learners and educators. However, LLMs, while effective at generating coherent responses, remain prone to pedagogical hallucinations (Ji, 2022) and may sometimes produce incorrect or biased explanations.

To mitigate these limitations, we adopt a neuro-symbolic approach, combining LLMs with KGs. This synergy allows us to: (1) anchor explanations in structured knowledge bases, (2) validate responses before delivering them to learners, and (3) adapt explanations based on detected emotions.

The proposed architecture relies on a structured explanation flow, combining neural reasoning (LLMs) and symbolic reasoning (KGs), as illustrated in Figure 1.

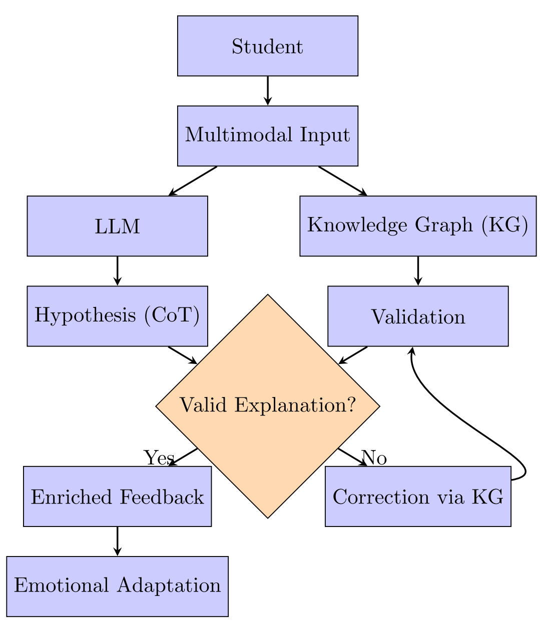

Fig. 1 Neuro-symbolic architecture for explainability in AI-powered education

This architecture operates through two complementary mechanisms: Knowledge Anchoring and Empathic Chain-of-Thought.

Knowledge anchoring

Knowledge Anchoring ensures that LLM-generated explanations are grounded in verifiable nodes within a Knowledge Graph, reducing the risk of pedagogical hallucinations. Each step of the reasoning process is linked to precise references within the KG, thereby enhancing semantic and pedagogical coherence (Hogan, 2020). For instance, MindfulTutor applies this technique to validate scientific concepts in chemistry using PubChem KG; when a student commits an error, the system identifies the corresponding node and delivers a corrected, evidence-based explanation.

Empathic chain-of-thought

Empathic Chain-of-Thought (Empathic CoT) extends standard reasoning mechanisms by dynamically adjusting explanations based on detected emotional signals–intonation, hesitation, response latency–from the learner (Poria, 2019). EduExplain exemplifies this mechanism: when a learner expresses frustration, the system revisits foundational concepts with softer, reassuring language; conversely, displays of confidence prompt advanced challenges that deepen understanding.

To quantify the impact of these mechanisms, we employ two complementary metrics: **PedagogyScore**, which measures the pedagogical quality of explanations through their coherence with verified KG references and logical consistency, and **StressClarity Index**, which quantifies the emotional impact of explanations via physiological measures (HRV, GSR) and validated psychometric assessments (Shih et al., 2024). OLMo Edu achieves high PedagogyScore ratings through its rigorous alignment with Wikidata, while MetaTutor demonstrates marked improvements in StressClarity Index through adaptive anxiety-reduction interventions.

Innovative Applications and Practical Implications

The integration of hybrid AI tutoring systems and collaborative knowledge-validation platforms has yielded demonstrable advances in personalized and emotion-aware education. Two exemplary cases illustrate these innovations:

MindfulTutor: Hybrid AI Tutoring System

MindfulTutor (MIT, 2024) represents a next-generation intelligent tutoring system that combines GPT-4, an affective Knowledge Graph (EmoKG), and real-time biometric sensors (heart rate variability, facial recognition) to deliver emotion-aware, multimodal instruction. The system adapts explanations dynamically in response to detected stress levels and cognitive load, employing PedagogyScore and StressClarity Index to personalize responses.

Empirical results demonstrate significant performance improvements across diverse learner profiles, as shown in Table 8 and visualized in Figure 2. Low-anxiety students improved from 70% to 85% comprehension; high-anxiety students exhibited the most dramatic gains, rising from 50% to 90%; and neurodivergent students improved from 55% to 88% comprehension. These gains underscore the pedagogical value of emotion-aware

Table 8 Impact of MindfulTutor on student comprehension across learner profiles

Student group	Standard AI tutoring (%)	MindfulTutor (%)
Low anxiety students	70	85
High anxiety students	50	90
Neurodivergent students	55	88

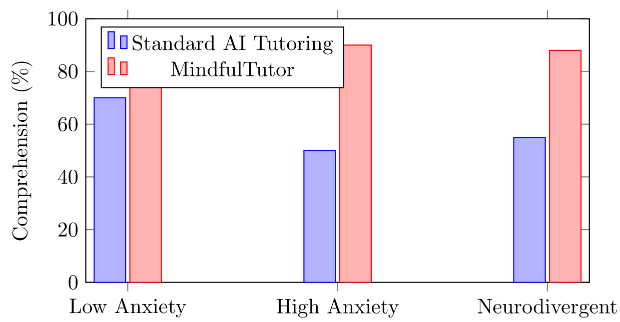

Fig. 2 Comparative student comprehension with MindfulTutor versus standard AI tutoring

Table 9 Explanation fidelity score improvements through educator review on EduExplain

Criterion	Without human review (%)	With human review (%)
Accuracy	72	91
Pedagogical clarity	68	88
Curriculum alignment	75	93

AI. A representative scenario illustrates the mechanism: when a student exhibits physiological stress indicators during calculus instruction, MindfulTutor detects these signals, consults EmoKG for emotion-adaptive strategies, reformulates explanations with increased scaffolding and reassuring language, and iteratively refines the learning trajectory based on ongoing physiological monitoring.

EduExplain: collaborative knowledge-validation platform

EduExplain operates as a decentralized, crowdsourced platform enabling educators to validate and refine AI-generated explanations, ensuring pedagogical alignment and trustworthiness. The platform leverages blockchain technology for transparent, auditable tracking of educator contributions and assigns an Explanation Fidelity Score (EFS) based on peer validation and alignment with verified knowledge sources.

Quantitative evaluation demonstrates substantial improvements when AI-generated content undergoes human review, as detailed in Table 9 and illustrated in Figure 3. Accuracy improves from 72% to 91%; pedagogical clarity rises from 68% to 88%; and curriculum alignment increases from 75% to 93%. These gains highlight the irreducible value of educator expertise in validating AI-generated explanations. A prototypical workflow: when a history teacher annotates an AI-generated explanation of the French

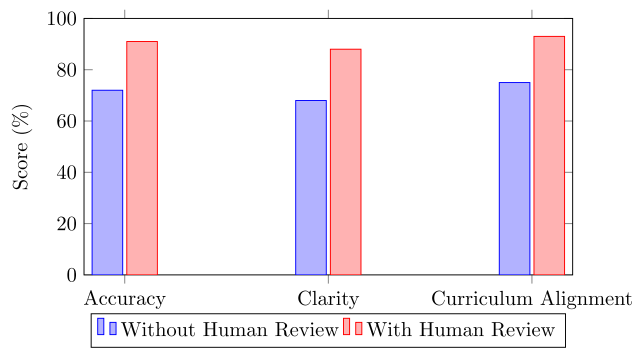

Fig. 3 Improvement in explanation fidelity metrics with educator validation

Table 10 Trade-offs between explanation simplification and learning outcomes

Explanation type	Cognitive load reduction	Concept retention	Student engagement
Simplified (basic terms, minimal details)	50%	60%	75%
Step-by-step (detailed breakdown)	30%	85%	90%
Deep reasoning (rich context, abstract concepts)	10%	92%	60%

Revolution, identifies factual inaccuracies, and submits corrections for peer validation, the refined explanation receives a high EFS, ensuring students encounter verified, pedagogically sound content.

These innovations collectively demonstrate that emotion-aware, hybrid AI systems–when complemented by human expertise–enable adaptive, personalized, and trustworthy educational experiences. The synthesis of automation and human judgment creates unprecedented opportunity for scaled pedagogical impact.

Ethical and technical challenges: critical tensions and mitigation strategies

Despite these advances, the integration of neuro-symbolic and affective AI into educational settings introduces profound ethical and technical tensions that demand systematic attention. Three fundamental challenges merit sustained investigation:

The explainability-cognition trade-Off

AI-driven tutoring systems dynamically simplify explanations for students experiencing cognitive overload. This raises a critical ethical question: *Should explanations be simplified when learners are stressed, even if doing so reduces pedagogical depth?*

Table 10 compares three explanation strategies and their differential impacts on learning outcomes. Simplified explanations achieve 50% cognitive load reduction but yield only 60% concept retention. Step-by-step breakdowns offer superior balance: 30% cognitive load reduction with 85% retention and 90% engagement. Deep reasoning explanations achieve highest retention (92%) but risk cognitive overwhelm (10% load reduction, 60% engagement). These results reveal a fundamental tension: cognitive accessibility and conceptual mastery are not automatically compatible. Responsive educational AI

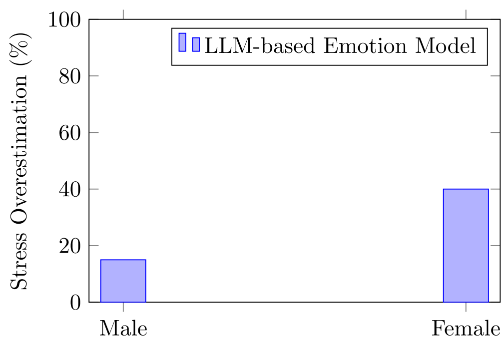

Fig. 4 Gender bias in AI-based stress prediction: overestimation in female versus male students

Table 11 Emotion detection accuracy: impact of biometric validation (EmoAnchor framework)

Emotion model	Accuracy without biometric	Accuracy with ECG validation
Traditional LLM	78%	–
Hybrid LLM-KG	85%	–
EmoAnchor (LLM + ECG)	76%	98%

must employ hierarchical explanation frameworks that maintain pedagogical rigor while accommodating learner cognitive state.

Affective recognition bias and equity

Affective AI models trained on demographically skewed datasets systematically misclassify emotional states across gender and cultural populations. Figure 4 reveals critical bias: AI stress detection overestimates stress by 40% in female students compared to 15% in male students. This misclassification perpetuates inequitable educational outcomes: female students receive unnecessary content simplification while male peers receive appropriately complex explanations. Such biases directly contravene educational equity principles and demand multimodal, demographically-representative training datasets.

Recent advances in emotion modeling, exemplified by EmoAnchor–an emotion validation framework integrating biometric data (ECG sensors)–demonstrate substantial bias mitigation (Table 11). EmoAnchor achieves 98% emotion detection accuracy when biometric validation is integrated, compared to 76% without such grounding. These improvements suggest that multimodal affective assessment, combining linguistic, behavioral, and physiological indicators, is essential for equitable AI-driven education.

Data privacy and personalization

Achieving meaningful personalization in AI tutoring requires granular collection of student data–behavioral patterns, performance histories, biometric signals, interaction logs. Yet centralized data architectures introduce substantial privacy risks and potential for misuse. Federated Explanation Training (FET) offers a privacy-preserving alternative by training models locally on individual devices, eliminating data transmission to centralized servers.

Table 12 contrasts centralized and federated approaches. While centralized AI enables moderate personalization at high privacy risk, federated AI maintains equivalent personalization while substantially reducing breach risk and preserving data autonomy. However, federated approaches entail higher computational costs, creating implementation challenges in resource-constrained contexts. Balancing privacy and personalization remains an open challenge requiring continued innovation in edge computing and privacy-preserving machine learning.

Future directions: towards ethical, explainable, and equitable AI in education

Addressing these systemic challenges necessitates coordinated advancement across multiple research and practice domains:

Optimization of adaptive explanation complexity

Future AI tutoring systems must employ hierarchical explanation frameworks that dynamically adjust semantic complexity, pedagogical depth, and scaffolding intensity in response to learner cognitive state, while maintaining conceptual integrity. This requires integration of cognitive load theory with machine learning-driven personalization, enabling just-in-time explanation refinement without introducing unnecessary oversimplification.

Multimodal affective assessment and bias mitigation

Advancing affective AI demands integration of linguistic, behavioral, and biometric indicators–facial expressions, vocal patterns, heart rate variability, galvanic skin response–into comprehensive emotion models. Training such models on demographically diverse, culturally representative datasets is essential for reducing gender, racial, and cultural bias. Standardization of affective ontologies and emotion taxonomy would facilitate cross-platform comparability and bias assessment.

Privacy-preserving personalization via federated learning

Expanding federated learning architectures for educational AI represents a critical priority. Future work should explore federated knowledge distillation, split learning, and differential privacy techniques to enable sophisticated personalization while preserving learner data autonomy. Such approaches require sustained collaboration between machine learning researchers, privacy advocates, educators, and policymakers.

Table 12 Privacy-personalization trade-offs: centralized versus Federated AI architectures

Criterion	Centralized AI	Federated AI (FET)
Data privacy	Low	High
Personalization	High	High
Risk of data breach	High	Low
Computational cost	Moderate	High
Adaptability to individual learners	Moderate	High

Emerging opportunities: explainable digital twins and evolving knowledge graphs

Two complementary technological directions offer particular promise:

Explainable Digital Twins for Personalized Learning Explainable Digital Twins (EDTs) simulate individual learner comprehension and emotional responses to instructional content, dynamically modeling cognitive processes and affective states in real-time. Unlike static user profiles, EDTs enable AI tutors to refine instructional strategies adaptively, integrating historical performance data, engagement patterns, and biometric signals. EDTs provide unprecedented transparency into AI-driven instructional decision-making, making pedagogical reasoning interpretable to students, educators, and parents.

Evolving Knowledge Graphs for Adaptive Learning Evolving Knowledge Graphs extend static KG paradigms by dynamically updating conceptual relationships in response to learner interactions, cognitive state indicators, and affective feedback. This approach enables identification of persistent misconceptions, refinement of pedagogical sequencing, and real-time restructuring of explanations to match evolving learner understanding. Integration of learner-generated annotations into evolving KGs creates collaborative, bidirectional learning environments where student input directly improves system performance.

Conclusion

While neuro-symbolic systems integrating LLMs, KGs, and affective computing have demonstrated substantial progress toward transparent and emotion-aware educational AI, realizing their full potential requires strategic investment in standardization, metacognitive architecture design, and ethical governance frameworks.

Standardization and interoperability

Future advancements in AI-powered education require establishing international benchmarks such as the ISO/IEC 23894-2 standard to ensure interoperability and consistency in the integration of emotional data within educational platforms. Standardized affective ontologies and emotion taxonomies would facilitate cross-platform comparability and enable large-scale deployment of equitable, validated systems.

Metacognitive and self-aware AI systems

Developing metacognitively-aware LLMs that incorporate internal self-explanation mechanisms represents a critical frontier. Such systems should assess and refine their own reasoning dynamically, enhancing pedagogical trust by enabling students to understand not just the "what" and "why," but also the system's own confidence and epistemic uncertainty. This self-aware capability positions AI as an interactive partner in learning rather than a mere content delivery mechanism.

Toward human-centered, equitable AI in education

By advancing toward these strategic directions–standardization, metacognitive transparency, privacy-preserving architectures, and ethical stewardship–AI in education can transition toward systems that are simultaneously explainable, adaptive, and equitable. This integrated approach positions educators and learners as active partners in shaping

AI systems that respect cognitive diversity, emotional well-being, and pedagogical integrity, ultimately fulfilling the promise of technology to enhance rather than diminish the human dimensions of education.

Author Contributions
All authors have read and approved the final manuscript.

Funding
No funding was obtained for this study.

Data Availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing financial or non-financial interests.

Received: 8 March 2025 Accepted: 24 November 2025
Published online: 20 January 2026

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